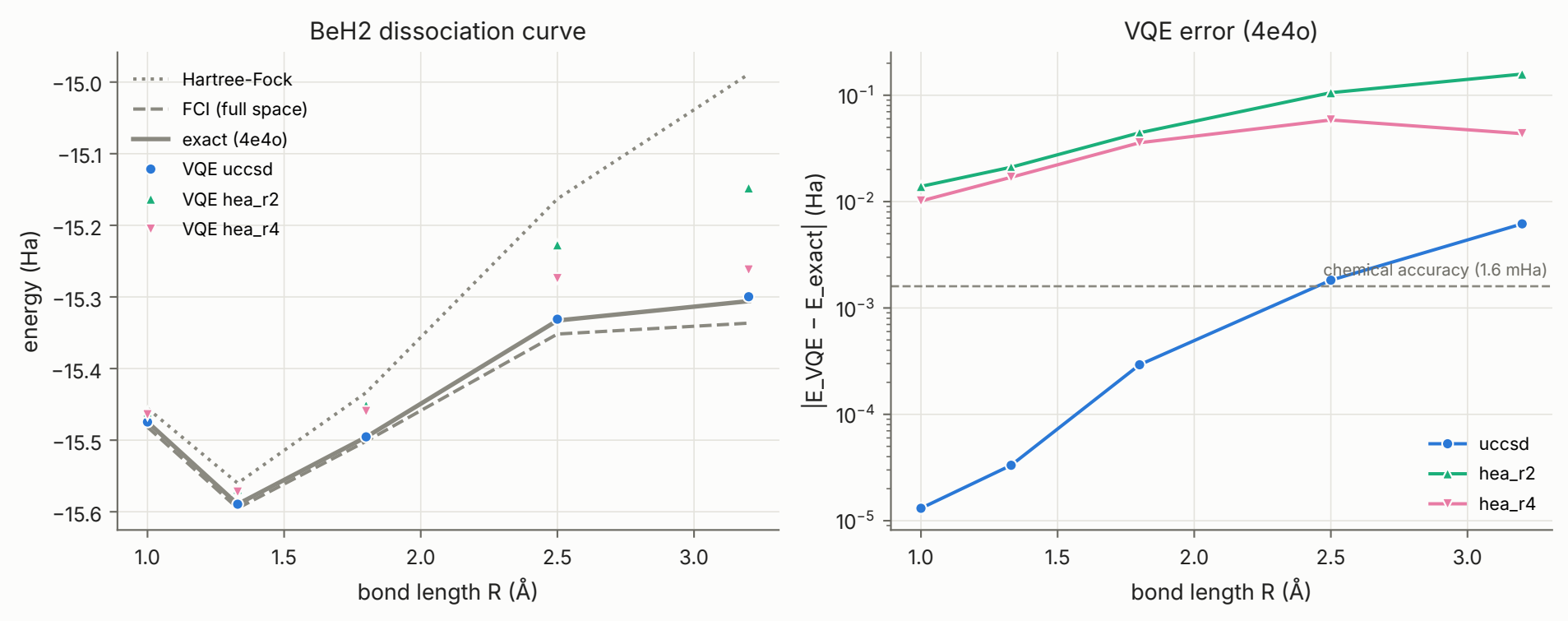
BeH2 dissociation curve
Hartree-Fock
FCI (full space)
exact (4e4o)
VQE uccsd
VQE hea_r2
VQE hea_r4
energy (Ha)
−15.0
−15.1
−15.2
−15.3
−15.4
−15.5
−15.6
1.0
1.5
2.0
2.5
3.0
bond length R (Å)
VQE error (4e4o)
|E_VQE − E_exact| (Ha)
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
chemical accuracy (1.6 mHa)
uccsd
hea_r2
hea_r4
1.0
1.5
2.0
2.5
3.0
bond length R (Å)